Questions

Q1.

A car collides with a barrier on a road.

The time of the collision is very short.

Explain **one** factor, other than the time of the collision, that would affect the force on the car in the collision.

Your explanation should refer to an equation in the Equation Booklet.

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(Total for question = 2 marks)

Q2.

Figure 12 is a diagram showing a rocket that is sent into space to try and change the path of a small asteroid.

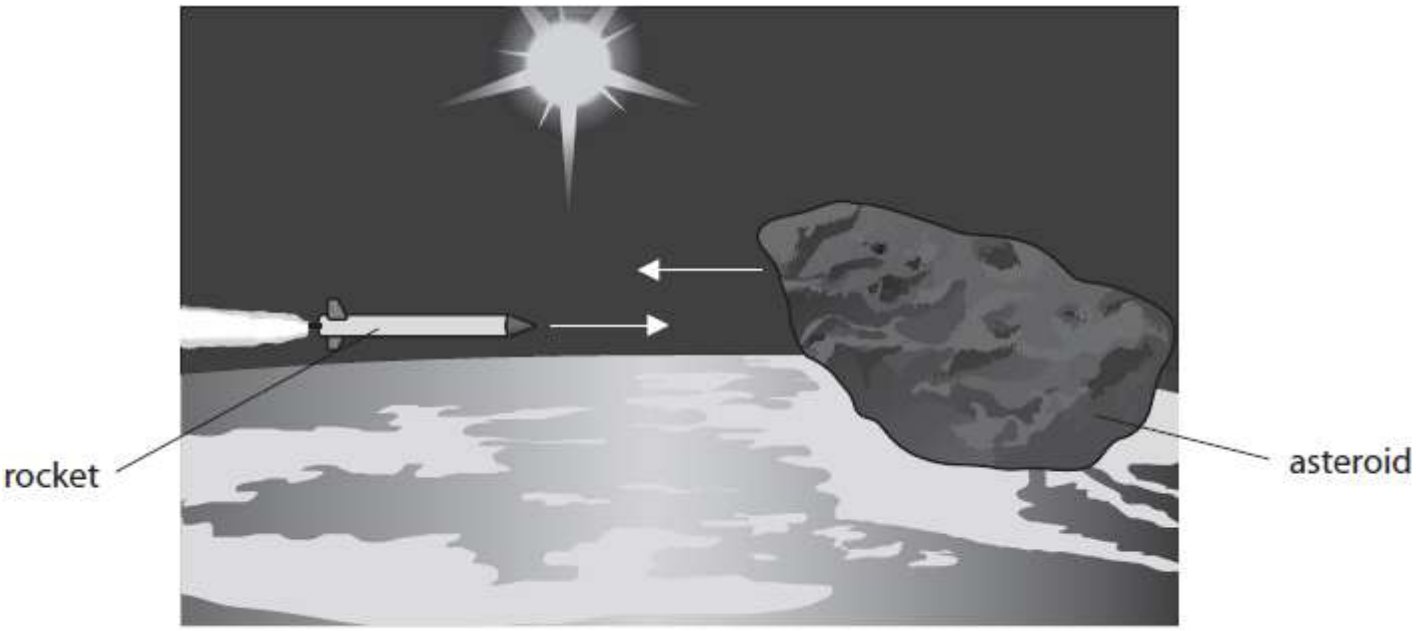

Figure 12

Newton's third law, when applied to the collision of the rocket and the asteroid as shown in Figure 12, can be stated as follows:

The force exerted by the rocket on the asteroid is equal and opposite to the force exerted by the asteroid on the rocket.

Explain how this statement links to the conservation of momentum in the collision.

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(Total for question = 4 marks)

Q3.

Figure 15 shows two trolleys, **P** and **Q**, moving at the same speed, v, directly towards each other.

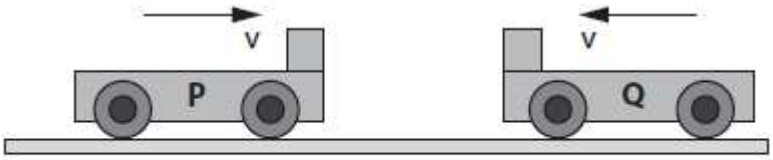

Figure 15

The trolleys have the same mass.

When the trolleys collide, they stick together and stop.

Explain how momentum and energy are both conserved in this collision.

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(Total for question = 6 marks)

Q4.

Figure 7 shows a submarine being propelled forward underwater.

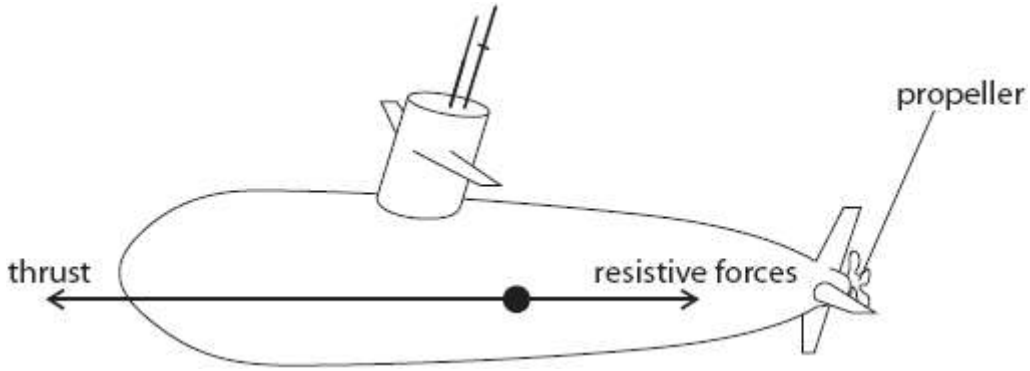

Figure 7

The thrust and the resistive forces are represented in magnitude and direction by the arrows in Figure 7.

* A submerged submarine is stationary.
The engines are put on maximum power.
The submarine moves forward.
The engines maintain maximum power.
The forces resisting the movement of the submarine increase as its speed increases.

The submarine remains horizontal. Explain how the forces acting on the submarine affect its acceleration up to and after it reaches its maximum possible speed.

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(Total for question = 6 marks)

Q5.

Shot-put is an Olympic event.
The shot is a heavy ball.
An athlete throws the shot as far as possible.

A sports scientist analyses an athlete's throw to help improve performance.

The scientist takes pictures of the athlete every 0.1 s during one throw.

Figure 6 shows the pictures of one throw.

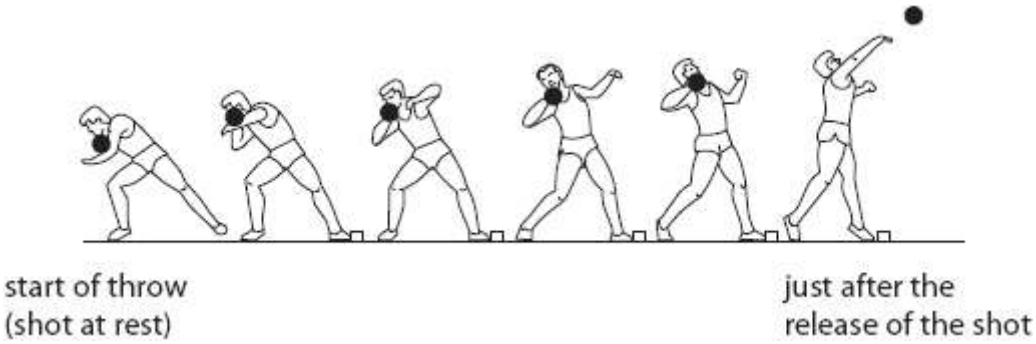

Figure 6

(i) Estimate the amount of time during the throw when the shot is in the athlete's hand.

(1)

time = ... s

(ii) Explain how the scientist could improve this method of analysing the throw.

(2)

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(iii) The average acceleration of the shot while in the athlete's hand is 20.6 m/s^2.

The mass of the shot is 7.26 kg.
Calculate the average force that the athlete applies to the shot during the throw.

(2)

force = ... N

(iv) In another throw, the shot is in the athlete's hand for 0.48 s.

The average acceleration during this time is 23 m/s^2.
Calculate the velocity of the shot as it leaves the athlete's hand.

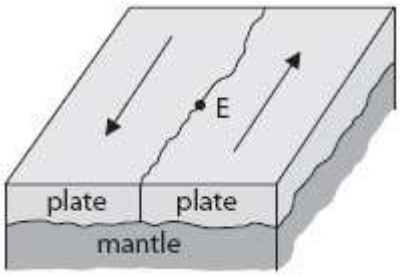
(3)

velocity = ... m/s

(Total for question = 8 marks)

Q6.

(a) The diagram shows part of the boundary between two tectonic plates.



(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

The plates are being steadily pushed in opposite directions by

(1)

☐ **A** convection currents in the mantle

☐ **B** reflection of waves from the Earth's core

☒ **C** tsunami waves in the ocean

☒ **D** volcanic eruptions on the surface

(ii) An earthquake occurs.
Its epicentre is at the place marked E on the diagram.
Describe what happens at the plate boundary to cause this earthquake.

(2)

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(b) The earthquake causes seismic waves.

(i) S waves are one type of seismic wave. They travel at 0.65 km/s.
There is a seismometer 80 km away from point E.
Show that it takes about 2 minutes for the S waves from the earthquake to reach the seismometer.

(2)

(ii) P waves are another type of seismic wave.
They travel about 10 times more quickly than S waves.

Describe how scientists can use seismometer records of P and S waves to locate the epicentre.

(3)

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(iii) Seismic waves have a frequency of about 15 Hz.
P waves have a much smaller amplitude than S waves.

Some people claim that animals can detect an earthquake before people are aware of it.
Suggest an explanation for this.

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(Total for Question is 10 marks)

Q7.

Figure 8 is a velocity/time graph for a lift moving upwards in a tall building.

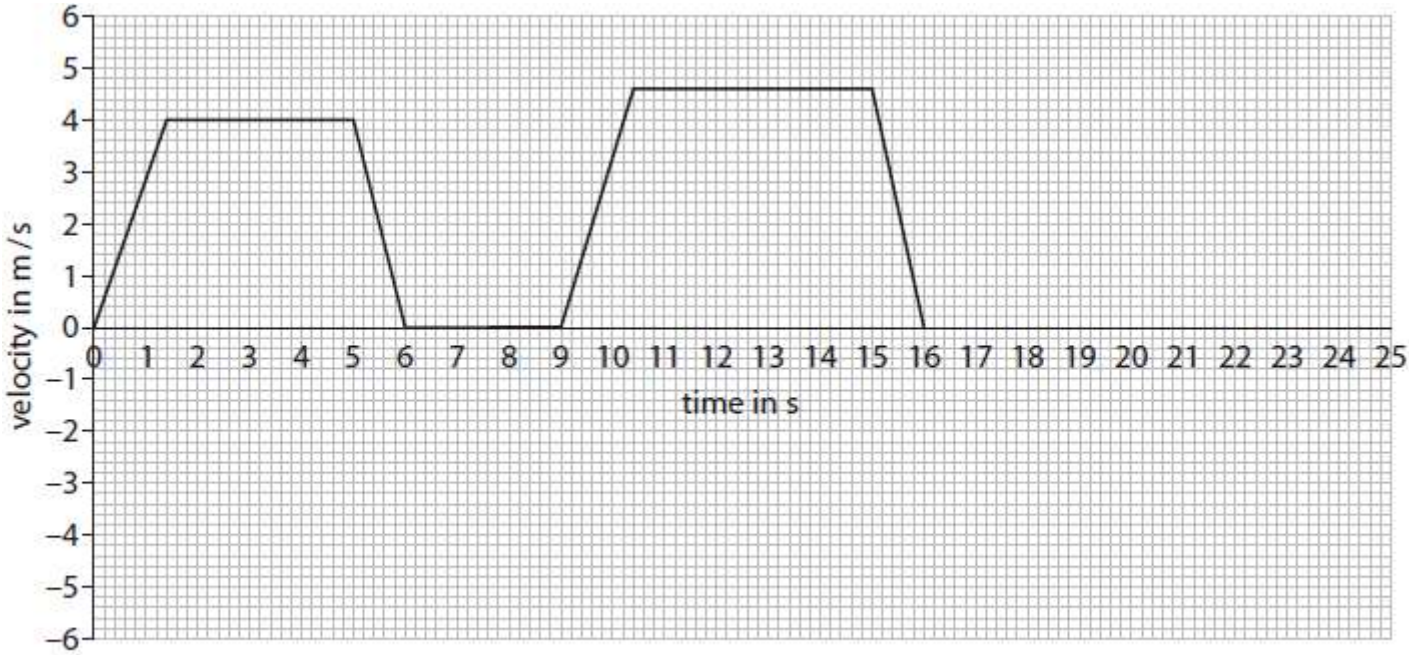

Figure 8

At 18 s, the lift starts to move downwards.

Sketch a line on the graph in Figure 8 to show the lift moving downwards after 18 s.

(1)

(Total for question = 1 mark)

Q8.

Describe how the student could extend the investigation to determine the average speed of the trolley as it rolls back down the track.

(3)

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(Total for question = 3 marks)

Q9.

A student lifts a toy car from a bench and places the toy car at the top of a slope as shown in Figure 2.

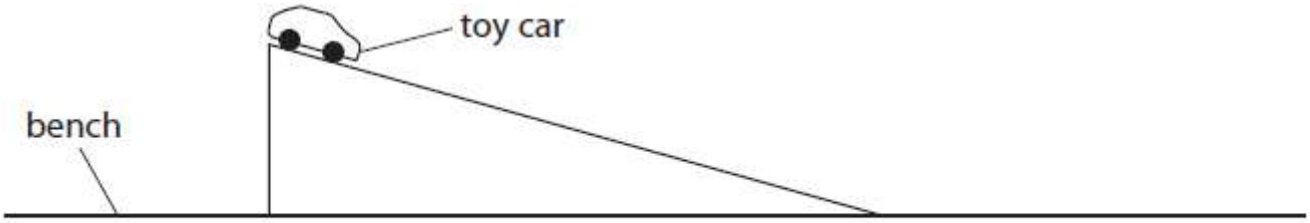

Figure 2

The student lets the toy car roll down the slope.

Describe how the student could find, by experiment, the speed of the toy car at the bottom of the slope.

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(Total for question = 4 marks)

Q10.

* Figure 14 shows two ice skaters during a sequence in their performance.

Figure 14

The man stays at the same place on the ice throughout the sequence.

At the start of the sequence, the woman is moving at a constant speed around the man while the man holds her arm.

After she has gone round the man several times, the man lets go of the woman's arm.
The sequence ends a few seconds later.

Explain the motion of the woman, in terms of the forces acting and the effects on her motion, for the whole sequence.

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(Total for question = 6 marks)

Q11.

Shot-put is an Olympic event.
The shot is a heavy ball.

An athlete throws the shot as far as possible.

A sports scientist analyses an athlete's throw to help improve performance.

The scientist examines the images to find ways of increasing the momentum of the shot when it leaves the athlete's hand without the athlete using any extra force.

The scientist advises the athlete to

lean further down at the start
and make his arm straight before he releases the shot.

Explain the scientific principles behind this advice.

(3)

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(Total for question = 3 marks)

Mark Scheme

Q1.

Question	Answer	Additional guidance	Mark
	explanation linking mass/(change of/initial/final) velocity/(change of) momentum (1) with a consistent equation linking chosen factor to force (1)	 accept speed/acceleration do not accept weight second marking point dependent on first marking point relevant equations are: $F = ma$ $F = \frac{mv - mu}{t}$ $F = \frac{\Delta p}{t}$ $F = \frac{m\Delta v}{t}$	**2** **AO2.1**

Q2.

Question number	Answer	Additional guidance	Mark
	An explanation linking any four from		(4) Exp
	force(s) associated with change(s) in momentum (use of Newton's second law) (1)	$F = \frac{(mv-mu)}{t}$ $F=ma$	
	detail of momentum changes, involving time (1)	$\frac{m_1 v_1 - m_1 u_1}{t} = -\frac{(m_2 v_2 - m_2 u_2)}{t}$	
	time of collision is same for both (1)		
	(therefore) momentum change is the same for both (1)		
	equal and opposite forces mean equal and opposite momentum changes (1)	with explanation leading to $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$ for both marks	
	(total) momentum before a collision = (total) momentum after collision (1)		
	(conservation of momentum requires) no external forces acting (1)		

Q3.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. momentum • momentum is product of mass and velocity • $p=mv$ • momentum is a vector quantity • momentum of P = mv • momentum of Q =-mv • total momentum before collision is zero • $mv_p + (-mv_q) = 0$ • momentum after collision must be zero • total momentum before = total momentum after	**6** **AO1.2** **AO2.2**
	energy • before collision each trolley has kinetic energy • $KE = \frac{1}{2}mv^2$ • total KE before collision is sum of KE of both trolleys • KE after collision is zero • KE transferred to thermal (store) • KE transferred by heating and by sound • energy dissipated to surroundings **Ignore references to equal and opposite forces and the laws of motion**	

Level	Mark	Descriptor
	0	No awardable content
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated fact(s) about the momentum and/or energy of trolleys, before or after collision	Possible candidate responses momentum before collision equals momentum after collision momentum is mass x velocity momentum is a vector energy cannot be created or destroyed $KE=\frac{1}{2}mv^2$
Level 2	3–4	Additional guidance limited description of conservation of momentum and limited description of conservation of energy OR detailed description of either conservation of energy or conservation of momentum	Possible candidate responses momentum/velocities of trolleys is equal and opposite before collision / zero before collision/zero after collision. trolleys have kinetic energy before collision OR the trolleys have kinetic energy before the collision which is transferred to thermal energy after the collision.
Level 3	5–6	Additional guidance detailed description of conservation of momentum before and after collision and limited description of conservation of energy OR detailed description of energy conservation and limited description of conservation of momentum	Possible candidate responses momentum of trolleys before collision is equal and opposite giving zero, and momentum is zero after collision so momentum is conserved energy is dissipated thermally after collision OR the trolleys have kinetic energy before the collision which is transferred to thermal energy after the collision. momentum of trolleys is equal and opposite before collision / zero before

			collision/zero after collision.

Q4.

Question Number	Answer	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 (6 marks)** **Stage 1** • (At the beginning) maximum force is applied • With zero resistive force(s) • And maximum acceleration **Stage 2** • As the speed increases the increasing resistive force results in a decreasing resultant forward force • And a decreasing acceleration • So velocity increases at a decreasing rate **Stage 3** • Eventually thrust = resistive force(s) • Then no resultant force forward and no acceleration • Submarine continues forward at a constant speed	**(6)**

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3–4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5–6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Q5.

Question number	Answer	Additional guidance	Mark
(i)	0.45 (s) (1)	Allow any value ≥ 0.4 and ≤ 0.5	**(1)**

Question number	Answer	Additional guidance	Mark
(ii)	An explanation that combines improvement of the experimental procedure (1 mark) and justification/reasoning which must be linked to the improvement (1 mark) • take pictures more frequently (1) • in order to determine exact time of the release. (1)	other responses may be acceptable	**(2)**

Question number	Answer	Additional guidance	Mark
(iii)	Substitution (1) F = 7.26 x 20.6 Evaluation (1) 150 (N)	Accept 149.6 (N) full marks will be awarded for correct numerical answer without working	(2)

Question number	Answer	Additional guidance	Mark
(iv)	Rearrangement (1) v = a x t Substitution (1) v = 23 x 0.48 Evaluation (1) 11 m/s	Accept 11.04(m/s) full marks will be awarded for correct numerical answer without working	(3)

Q6.

	Answer	Acceptable answers	Mark
(ai)	A		(1)
(aii)	A description linking plates move / slip / separate (relative to each other) (1) sudden (release of energy) (1)	plate rubs against each other friction between plates plate boundary shifts jerk / jolt	(2)
(bi)	substitution (1) 0.65 = 80 / t transposition (1) t = 80 / 0.65 (123 seconds)	transposition and substitution can be in either order . Allow reverse calculations eg speed = 80/120 (1) = 0.67 (about 0.65) (1) or distance = 0.65 × 120 (1) = 78 km (about 80) (1).	(2)

(bii)	A description linking any three detection of arrival of P and S waves (1) measurement of difference in arrival times (1) calculation of distance (from epicentre to station) (1) triangulation/using three / several stations (1)	Reward suitable labelled diagram	**(3)**
(b)(iii)	A suggestion including any two of the following Infrasound (1) some animals can hear waves below human frequency range / 20 Hz (1) they could hear P waves arriving before the (stronger) S waves arrive (1)	Some animals have greater audio / tactile sensitivity than humans	**(2)**

Q7.

	Answer	Additional guidance	Mark
	graph continued below time axis, starting at 18 s	do not accept vertical line do not accept line extending to left of 18 s	**(1)** **AO3**

Q8.

Question Number	Answer	Additional guidance	Mark
	A description including:		**(3)**
	measure appropriate distance (1)	e.g. distance along runway from max height to P	
	measure appropriate time (1)	e.g. start the watch when trolley stops stop the watch when trolley hits spring	
	use (average) speed = $\frac{\text{distance}}{\text{time}}$ (1)	accept $s = \frac{d}{t}$	

Q9.

Question number	Answer	Additional guidance	Mark
	A description to include:		**(4)**
	measurement of (relevant) distance (1)	one of distance down slope or distance along bench or length of toy car/card	
	measurement of (relevant) time (1)	'record the distance the car travels and time it' scores 2 marks	
	use of speed = $\frac{\text{distance}}{\text{time}}$ (1)		
	detail (1)	for example: speed down slope × 2 mark distance along bench use a light gate speed gun at the bottom of the slope repeating AND averaging	

Q10.

Question number	Indicative content	Mark
*	The indicative content below is not prescriptive and candidates are not required to include all of the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **A01 (6 marks)** • force needed to keep an object moving in a circle • when moving in a circle, direction of velocity changes • must be an acceleration • moving in a straight line with no resultant force at constant velocity **A02 (6 marks)** • the woman changing direction while circling the man • she is changing velocity (but not changing speed) • therefore she is accelerating • this requires a force towards the centre of her orbit • this is a centripetal force • when the man releases the woman, the centripetal force ceases • there is no resultant force on the woman (if friction from the ice can be ignored) • the woman therefore continue in a straight line • she is now travelling at a constant velocity	**(6)**

Level	Mark	Descriptor
	0	No awardable content
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

Q11.

Question number	Answer	Additional guidance	Mark
	An explanation that combines up to 3 points of application of knowledge and reasoning/justification • Momentum increased if final velocity can be increased (1) • Distance (while in athlete's hand) is greater (1) • Time whilst subject to force is longer (1) • using the equation F = (mv-mu)/t (1)	Ignore references to shot after it has left the athletes hand accelerating for a longer time use of v = u + at or use of v^2-u^2 = 2xaxd	**(3)**